Recurrent Latent-Space Reasoning: Watching a Model Think Without Words

Team root16 — DataForge 2026 × Pathway × rime — Pathway Track — One-Page Concept Summary

The claim (falsifiable): A model can improve an answer by repeatedly transforming a fixed-size hidden state a variable number of times — without ever converting an intermediate step into a word or token. If this were false, giving a model more “thinking passes” over the same hidden state would do nothing: accuracy would sit flat regardless of iteration count.

Why this matters now

Most people's picture of “AI reasoning” is chain-of-thought (CoT): the model writes intermediate steps in natural language, and every step costs real output tokens — expensive, and it forces computation that may not be naturally verbal into words anyway. An alternative line of work asks what happens if a model instead reasons by repeatedly transforming a hidden state, refining it silently and decoding only once, at the end. Hao et al. (2024, arXiv:2412.06769) first trained an autoregressive LLM to reason this way, in continuous latent space instead of words. Geiping et al. (2025, arXiv:2502.05171) showed a recurrent-depth architecture that scales test-time compute by looping a shared block over a latent state rather than growing the model or the token count. Saunshi et al. (2025, arXiv:2502.17416) gave the idea theoretical footing, proving looped transformers can simulate bounded chain-of-thought computation without adding parameters per step. Pathway's BDH-CQ system is a recent, concrete instance of this same mechanism, which is why it anchors this explainer.

What the artifact does

index.html is a small, real, running toy model, not an animation. It shows two demonstration grids encoding a hidden rule (a positional shift plus a 4-color permutation) and holds a fixed hypothesis space of 54 candidates, scored by how well each reproduces the demonstrations. A reasoning-effort dial (0 / LOW / MEDIUM / HIGH → 0, 1, 4, 14 iterations, or an exact 0–20 slider) controls how many times the belief over those 54 hypotheses is refined: evidence accumulates linearly with iteration count, then passes through a softmax, so more iterations genuinely sharpen belief toward the best-supported rule. The top hypotheses are drawn live as bars — the belief only ever exists as numbers, never as text. The current best hypothesis is then applied to a held-out query grid and shown beside the true answer.

Running the falsification test: at 0 iterations, accuracy sits at chance (25%, since belief is uniform over 54 hypotheses). At HIGH, in the included worked example, one iteration already suffices to single out the correct rule; accuracy saturates at 100%, while belief confidence keeps sharpening (residual belief error decays exponentially) for many further iterations even after the answer itself stops changing — the model growing more confident, not just more accurate, as it “thinks” longer.

Where BDH and BDH-CQ fit

The toy model is a teaching device, not a reproduction of any Pathway system — but its mechanism (fixed-size state, refined over passes, never decoded mid-computation) is the same shape BDH and BDH-CQ implement at scale. BDH (Kosowski et al., 2025, arXiv:2509.26507) gives a sequence model a persistent, synapse-level state: attention emerges from local neuron-to-neuron interactions on a sparse, scale-free graph with Hebbian-style updates, yielding a fixed-size, continuously-updatable state instead of a growing key-value cache. BDH-CQ (Engdahl et al., 2026, arXiv:2608.09888) uses that state for two jobs: a contextual memory continuously updated by demonstrations (no weight changes — analogous to our two demo grids), and a separate latent workspace transformed repeatedly to solve a query (analogous to our effort dial). No backward pass or parameter update happens at inference; adaptation lives entirely in recurrent state. Reported figures (developer-reported): a 150M-parameter BDH-CQ configuration reaches 29.5% pass@2 / 24.25% pass@1 on public ARC-AGI-1, at ≈$0.00070/task. Effort tracks accuracy the same direction our dial does: LOW 21% → MEDIUM 27% → HIGH 29.5% pass@2.

What this doesn't prove, and a real limitation

More latent iterations is not a bigger or smarter model: turning the dial up gives the existing belief state more passes over the same fixed evidence, not new parameters or knowledge — the same is true of BDH-CQ's contextual-memory update, which is not a weight update. BDH-CQ's own controlled studies show strong extrapolation on boundary propagation, copying, and dense mappings, but performance drops sharply on long ordering sequences, deep nesting, conditional rule selection, and composing multiple transformations together (color-swap composed with relocation was solved 0/72 times). Our toy demo deliberately uses a small, disambiguated hypothesis space that always converges cleanly — the real system does not, and that gap is worth sitting with rather than smoothing over. Separately, its public ARC-AGI-1 audit was conducted by co-authors of the BDH-CQ paper at partner institutions: a genuine black-box audit without weight access, but not a fully external third-party reproduction — labeled here as such rather than as an unqualified “independent evaluation.”

Primary sources: Hao et al. 2024 (arXiv:2412.06769); Geiping et al. 2025 (arXiv:2502.05171); Saunshi et al. 2025 (arXiv:2502.17416); Kosowski et al. 2025 (arXiv:2509.26507); Engdahl et al. 2026 (arXiv:2608.09888). Full citation list with placement guidance: CITATIONS.md.